Imran Jahid Khan

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Research Interests

Security and Resilience of Cyber-Physical Systems • Sensor Attacks, Anomaly Detection, and Recovery • Robust Reinforcement Learning for Control • Cloud-Controlled and Medical CPS • Blockchain-Based Integrity Verification

Education

Tuskegee University, Alabama, USA *Aug 2025 – May 2027 (expected)*
M.S. in Information Systems and Computer Security
Advisor: Dr. Mohammad Arafatur Rahman
Relevant Coursework: Programming Language (C++); Information Security; Network Security and Management; Data Networks & Cloud Computing; Data Mining & Machine Learning; Statistics with R; Data Analytics; Big Data Analytics; Database Systems.

Bangladesh University of Engineering & Technology (BUET), Dhaka *2011 – 2016*
B.Sc. in Mechanical Engineering
Thesis: Wind Turbine Blade Design for Optimum Power Output.
Relevant Coursework: Computer Programming Language (C); Calculus; Linear Algebra; Ordinary & Partial Differential Equations; Complex Variables & Fourier Analysis; Numerical Analysis.

Research Experience

Graduate Research Assistant *Aug 2025 – Present*
Dept. of Computer Science, Tuskegee University • Advisor: Dr. Mohammad Arafatur Rahman

- Built a cloud-controlled Q-learning CartPole testbed; showed that low-variance observation noise interacts non-linearly with environmental stress (inclined surface), cutting mean survival time at the steepest incline from 4.82 s to 1.44 s [2].
- Evaluated six DQN training and reconstruction strategies across 240,000 episode records under Gaussian noise and randomized replay; under all-state replay, LSTM reconstruction had the highest mean survival (7.46 s), beating mixed-attack-aware DQN in all five runs; its edge over recurrent DQN was not significant [7].
- Combined a PPO controller, an LSTM-autoencoder detector, and a private blockchain ledger that detects replay deterministically and anchors recovery to verified clean states; restored 95.7% of clean return on LunarLander, while recovery from an unverified local buffer cut CartPole replay return from 324.5 to 69.1 [3].
- Showed that detect-then-recover fails under CGM attacks on an RL artificial pancreas (exact-match replay detection never fired; the windowed detector fired on 62–94% of clean-reading steps); designed detector-free residual-LSTM reconstruction with a safety shield that lowered hypoglycemia risk (LBGI) under replay, drift, and FDI [4].
- Co-authored studies of classical vs. quantum RL controllers [5, 8], DQN robustness to Gaussian sensor noise [9], PPO-based adaptive sampling for multi-plant cloud CPS [10], LSTM-based integrity-attack detection [1], and AI-assisted NMR metabolomics evidence ranking [6].

Publications

PUBLISHED CONFERENCE PAPERS

† presenting author

[1] M. Rahman, **I. J. Khan**, and F. Wu, "Detection of Integrity Attacks in Cloud-Based Cyber-Physical Systems (CBCPS) Using Long Short-Term Memory (LSTM) Models," in *Proc. 2026 9th International Conference on Information and Computer Technologies (ICICT)*, pp. 533–538, 2026.

[2] **I. J. Khan**, M. Hossain, M. Rahman, and F. Wu, "Resilience Analysis of Cloud-Controlled Cyber-Physical Systems Using a Reinforcement Learning CartPole Testbed," in *Proc. 2026 IEEE World AI IoT Congress (AIIoT)*, pp. 773–779, 2026.

ACCEPTED CONFERENCE / WORKSHOP PAPERS

[3] **I. J. Khan**†, M. Hossain, M. Rahman, and F. Wu, "Blockchain-Anchored Detection and Recovery for Resilient Reinforcement Learning-Based Cyber-Physical Systems," *7th Annual National HBCU Blockchain, FinTech, and AI Conference*, National FinTech Center, Morgan State University, 2026.

[4] **I. J. Khan**†, S. Tahmim, M. Hossain, M. Rahman, and F. Wu, "Reconstruction-Assisted Resilient Reinforcement Learning for the Artificial Pancreas under CGM Sensor Attacks," *Proc. IEEE International Conference on E-health Networking, Application & Services (HealthCom) Workshops, ISTHS Workshop*, 2026.

[5] M. Hossain, **I. J. Khan**†, M. Rahman, and F. Wu, "Evaluating Classical and Quantum Reinforcement Learning Controllers under Meal-Information Uncertainty in Artificial Pancreas Systems," *Proc. IEEE HealthCom Workshops, ISTHS Workshop*, 2026.

[6] S. Tahmim, **I. J. Khan**, and W. E. Collier, "An Auditable Evidence-Ranking Framework for AI-Assisted NMR Metabolomics in Breast Cancer Research," *Proc. IEEE HealthCom Workshops, ISTHS Workshop*, 2026.

[7] **I. J. Khan**, M. Hossain, M. Rahman, and F. Wu, "Reconstruction-Assisted DQN for Resilient Cloud-Controlled Cyber-Physical Systems against Data Integrity Attacks," *2026 IEEE International Conference on Big Data (BigData), 2nd Special Session on Cybersecurity and Telecommunication in the Era of AI*, 2026.

SUBMITTED / UNDER REVIEW

[8] M. Hossain, **I. J. Khan**, M. Rahman, and F. Wu, "Evaluating Learning Efficiency and Computational Trade-Offs in Classical and Quantum Reinforcement Learning Using a CartPole Testbed," submitted to *IEEE BigData 2026*.

[9] M. Rahman, **I. J. Khan**, M. Hossain, and F. Wu, "Evaluating DQN-Based Reinforcement Learning Robustness to Gaussian Sensor Noise in Cyber-Physical Systems: Training, Reconstruction, and Reward Design," submitted to *IJACSA*.

[10] M. Rahman, S. Nazrul, **I. J. Khan**, M. Hossain, and F. Wu, "Centralized PPO-Based Adaptive Sampling and Network Resource Management for Multi-Plant Cloud-Based Cyber-Physical Systems (CBCPS)," submitted to *IJACSA*.

Presentations

- **Oral, Nov 8–10, 2026 (scheduled):** paper [3], 7th Annual National HBCU Blockchain, FinTech, and AI Conference (National FinTech Center, Morgan State University), Nashville, TN.
- **Oral, Oct 2026:** papers [4] and [5], ISTHS Workshop, IEEE HealthCom 2026.
- **Poster, 2025:** "Evaluating Q-Learning Rewards in Noisy Cloud-Based Cyber-Physical Systems." *AWS-MLU AI/ML Fall Teaching and Research Symposium & Student Poster Session*.

Honors & Scholarships

- **Full Scholarship (Graduate Assistantship)**, M.S. program, Tuskegee University, 2025
- **Japan Human Resources Development Scholarship (JDS)**, Government of Japan, Hiroshima University, 2025 *(declined)*
- **Full Scholarship**, M.A. in Policy Economics, Williams College, 2025 *(declined)*
- **Bangladesh Civil Service (Administration) Cadre**, 37th BCS Examination: merit position 89 of 306 appointees from about 243,000 candidates
- **Prime Minister's Fellowship**, Government of Bangladesh, 2024
- **Foreign Master's Scholarship**, "Capacity Building of BCS Administration Academy (2nd Revised)" Project, Government of Bangladesh, 2024 *(declined)*

Teaching Experience

Graduate Teaching Assistant *Aug 2025 – Dec 2025*
Introduction to Computer Science, Dept. of Computer Science, Tuskegee University

Professional Experience (Bangladesh Civil Service)

Deputy Director (Senior Assistant Secretary) *Sep 2024 – Jul 2025*
Export Promotion Bureau, Ministry of Commerce

- Produced data-informed export-policy recommendations and worked with stakeholders to digitize trade-facilitation processes.

Assistant Commissioner (Land) & Executive Magistrate *Nov 2022 – Aug 2024*
Upazila Land Office, Patuakhali & Bhola, Ministry of Land

- Managed digital land records and reporting to support land administration; adjudicated land disputes and coordinated field operations.

Assistant Commissioner & Executive Magistrate *Apr 2019 – Nov 2022*
Office of the Deputy Commissioner, Cox's Bazar, Ministry of Public Administration

- Coordinated disaster-relief and COVID-19 response using ICT; administered public budgets, personnel, and electronic procurement.

Technical Skills

Programming & Tools: Python, C/C++, R, SQL, MATLAB; Git/GitHub, Linux, LaTeX
ML / RL: PyTorch; Q-learning, DQN, PPO; LSTM and LSTM Autoencoders; Anomaly Detection; State Reconstruction
Simulation: Gymnasium (CartPole, LunarLander), Simglucose (artificial-pancreas simulator)

Standardized Tests

GRE: 322 (Quantitative 167), Dec 2024 **IELTS Academic:** 7.0, Jan 2024